Observer Quality as a Resource Constraint in Quantum Darwinism: Optimal Decoding, ε-Approximate Spectrum Broadcast Structure, and a Central-Spin Worked Example

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Abstract

Quantum Darwinism (QD) and Spectrum Broadcast Structure (SBS) formalize a mechanism by which many observers can learn the same classical pointer value of a system by measuring different fragments of its environment. Most QD analyses treat the observer as idealized: unlimited access, perfect calibration, and no temporal constraints. This paper introduces an explicit *observer quality* parameterization and propagates it through redundancy and objectivity statements in a way that remains stable under an explicit ε-SBS trace-distance error model.

Our main technical contributions are: (i) a decoder-level, tight (Chernoff-optimal) sample-complexity characterization for the number of environmental fragments needed to infer a pointer value under a calibrated observer model, (ii) a data-processing theorem showing that calibration noise (modeled as a CPTP channel) degrades the quantum Chernoff exponent, and (iii) an ε-robustness theorem upgrading ideal-SBS sample-complexity bounds to approximate-SBS states with additive error control.

To address "no concrete physics" concerns, we provide a fully worked open-system example: a central-spin pure-dephasing model whose conditional fragment states can be computed analytically, yielding explicit formulas for the observer parameters (R_O, C_O, τ_O) as functions of couplings, readout noise, and acquisition time. We also highlight a sharp performance gap between collective (coherent) decoding and product-measurement decoding for pure-state records, directly linking *quantum memory horizon* to redundancy.

1 Introduction

Quantum Darwinism proposes that classical objectivity arises when information about a preferred pointer observable of a system is redundantly encoded in the environment [1–3]. A complementary structural characterization is Spectrum Broadcast Structure [4, 5], and strong-QD variants connect entropic and geometric criteria [5, 6].

A persistent modeling gap is that *observers are usually treated as ideal*. Yet in realistic experiments or cognitive settings, an observer may: (i) access only a fraction of environment fragments, (ii) have imperfect measurement calibration, and (iii) be limited by a temporal integration

horizon (finite memory or pointer stability). There is substantial prior work on non-idealities in the *environment* (e.g., "hazy" environments [7]) and on generic emergence of objectivity [8, 9], but the observer itself is rarely parameterized at the same level of explicitness.

Scope of this paper. We restrict to a single core task: *observer-relative redundancy/objectivity statements* in QD/SBS when the observer is explicitly modeled by a quality triple

$$Q_O = (R_O, \Lambda_O, \tau_O),$$

where R_O is an access fraction, Λ_O is a calibration channel applied to each accessed fragment prior to readout, and τ_O is an acquisition or memory horizon. For interpretability (and to align with existing QD language) we also extract a scalar *calibration index* C_O from Λ_O in the binary case.

Why an ε-SBS error model. Many QD/SBS derivations assume conditional i.i.d. record models. Instead, we start from an explicit trace-distance approximation: the actual joint state $\rho_{S\mathcal{E}}$ is within ε in trace distance of an *ideal* SBS state $\sigma_{S\mathcal{E}}$. We then push sample-complexity statements through this approximation using data processing and decision-theoretic continuity bounds.

2 Related work and positioning

This paper builds on three strands:

1. **QD and SBS foundations.** The environment-as-witness framing originates in Ollivier et al. [1] and is synthesized by Zurek [2], Blume-Kohout and Zurek [3]. SBS provides a structural notion of objectivity [4], and a unifying review is Korbicz [5]. Strong-QD links entropic and structural conditions [6].

2. **Non-ideal environments and fragment size.** "Hazy" environments reduce redundancy and record capacity [7]. Photon-scattering models exhibit huge redundancy in everyday environments [10, 11].

3. **Hypothesis testing and optimal decoding.** Binary quantum hypothesis testing is classical material [12, 13]. The quantum Chernoff bound characterizes the optimal asymptotic error exponent [14, 15]. These tools are natural for QD because "learning the pointer value" is precisely a discrimination task between conditional fragment states.

We also connect to recent resource-theoretic and operational approaches emphasizing *accessible information* rather than mutual information [16].

3 Setup: SBS, ε-SBS, and observer quality

3.1 Spectrum Broadcast Structure and its approximation

Let S be the system and $\mathcal{E} = \mathcal{E}_1 \otimes \cdots \otimes \mathcal{E}_N$ a decomposition of the environment into fragments. We focus on a classical pointer random variable X taking values in a finite alphabet $\mathcal{X}$ (often $\mathcal{X} = \{0, 1\}$).

Definition 1 (SBS state)**.** A joint state $\sigma_{S\mathcal{E}}$ has *spectrum broadcast structure* (SBS) with respect to pointer basis $\{|x\rangle\}_{x\in\mathcal{X}}$ if it can be written as

$$\sigma_{S\mathcal{E}} = \sum_{x\in\mathcal{X}} p_x \, |x\rangle\langle x|_S \otimes \bigotimes_{k=1}^{N} \sigma_{\mathcal{E}_k}^{(x)}, \tag{1}$$

where p is a probability distribution and, for each fragment k, the conditional states $\{\sigma_{\mathcal{E}_k}^{(x)}\}_{x\in\mathcal{X}}$ have mutually orthogonal supports (perfect distinguishability) [4, 5].

Definition 2 (ε-SBS)**.** A state $\rho_{S\mathcal{E}}$ is *ε-SBS* if there exists an SBS state $\sigma_{S\mathcal{E}}$ such that

$$D_{\mathrm{tr}}(\rho_{S\mathcal{E}}, \sigma_{S\mathcal{E}}) \le \varepsilon, \qquad D_{\mathrm{tr}}(\rho,\sigma) := \tfrac{1}{2}\left\|\rho-\sigma\right\|_1 .$$

3.2 Observer quality

An observer interacts only with a subset of fragments and through an imperfect instrument.

Definition 3 (Observer quality)**.** An observer is specified by a quality triple

$$Q_O = (R_O, \Lambda_O, \tau_O),$$

where:

1. $R_O \in [0,1]$ is an *access fraction*: out of N fragments, O can access at most $m_{\max} = \lfloor R_O N \rfloor$ fragments in a given observational episode;
2. Λ_O is a CPTP map acting on each accessed fragment, representing calibration noise, coarse-graining, or any pre-measurement processing;
3. τ_O is a temporal horizon (memory, stability, or acquisition time). In the central-spin model (Sec. 7), it is mapped to an acquisition time budget and a pointer-stability constraint.

When $\mathcal{X} = \{0,1\}$, we can compress Λ_O to a scalar "single-fragment calibration" index.

Definition 4 (Binary calibration index)**.** Let ρ_0, ρ_1 be the two conditional states of a single accessed fragment after Λ_O, and assume equal priors. Define

$$C_O := 1 - P_e^\star(\rho_0,\rho_1), \qquad P_e^\star(\rho_0,\rho_1) = \tfrac{1}{2}\left(1 - \tfrac{1}{2}\left\|\rho_0-\rho_1\right\|_1\right), \tag{2}$$

where $P_e^\star$ is the Helstrom-optimal single-copy Bayes error [12, 13]. Then $C_O \in [\frac{1}{2}, 1]$.

4 Optimal decoding and Chernoff sample complexity

4.1 Binary hypothesis testing on fragments

Fix an SBS reference state $\sigma_{S\mathcal{E}}$ of the form (1) with $\mathcal{X} = \{0,1\}$. For any fragment set $A \subseteq \{1,\dots,N\}$, let

$$\sigma_A^{(x)} := \bigotimes_{k\in A} \sigma_{\mathcal{E}_k}^{(x)}$$

denote the conditional state of the accessible environment.

The observer first applies Λ_O to each accessed fragment and then performs an optimal measurement to decide x. This induces a binary quantum hypothesis testing problem:

$$H_0 : \ \rho_A = \bigotimes_{k\in A} \Lambda_O(\sigma_{\mathcal{E}_k}^{(0)}), \qquad H_1 : \ \rho_A = \bigotimes_{k\in A} \Lambda_O(\sigma_{\mathcal{E}_k}^{(1)}).$$

4.2 Quantum Chernoff bound and tight redundancy scaling

Define the (binary) quantum Chernoff coefficient for a pair of states ρ, σ:

$$Q(\rho, \sigma) := \min_{s \in [0,1]} \operatorname{Tr}\left[\rho^s \sigma^{1-s}\right],$$

and the associated Chernoff exponent

$$\xi(\rho, \sigma) := -\log Q(\rho, \sigma).$$

The quantum Chernoff bound states that for discriminating $\rho^{\otimes n}$ vs $\sigma^{\otimes n}$ with equal priors, the optimal error decays with exponent $\xi(\rho, \sigma)$ [14, 15].

We use a finite-n inequality that is convenient for sample-complexity statements.

Theorem 1 (Finite-n Chernoff upper bound)**.** *Let ρ, σ be density operators and consider equal priors on $H_0 : \rho$ and $H_1 : \sigma$. Then the optimal Bayes error obeys*

$$P_e^\star(\rho, \sigma) \leq \tfrac{1}{2}\, Q(\rho, \sigma). \tag{3}$$

Consequently, for product states over a fragment set A,

$$P_e^\star\left(\bigotimes_{k \in A} \rho_k, \bigotimes_{k \in A} \sigma_k\right) \leq \tfrac{1}{2} \min_{s \in [0,1]} \prod_{k \in A} \operatorname{Tr}\left[\rho_k^s \sigma_k^{1-s}\right].$$

Proof. Inequality (3) is a standard consequence of the Audenaert et al. bound and is commonly stated as $P_e^\star(\rho, \sigma) \leq \frac{1}{2} \min_{s \in [0,1]} \operatorname{Tr}[\rho^s \sigma^{1-s}]$ [14]. The product form follows from multiplicativity of the trace under tensor products. □

Definition 5 (Observer-effective Chernoff exponent on a fragment set)**.** For an observer O and fragment set A, define

$$\xi_O(A) := \max_{s \in [0,1]} \left(-\sum_{k \in A} \log \operatorname{Tr}\Big[\Lambda_O(\sigma_{\mathcal{E}_k}^{(0)})^s\, \Lambda_O(\sigma_{\mathcal{E}_k}^{(1)})^{1-s}\Big] \right). \tag{4}$$

Proposition 1 (Access threshold as an information budget)**.** *Fix an error target $\delta \in (0, \frac{1}{2})$ and an SBS reference experiment. If an observer can access at most $m_{\max} = \lfloor R_O N \rfloor$ fragments, then a sufficient condition for δ-decoding of the pointer value is the existence of a set A with $|A| \leq m_{\max}$ such that*

$$\xi_O(A) \geq \log\left(\frac{1}{2\delta}\right).$$

Proof. By Theorem 1, $P_e^\star(A) \leq \frac{1}{2} \exp(-\xi_O(A))$. If $\xi_O(A) \geq \log(1/(2\delta))$, then $P_e^\star(A) \leq \delta$. □

4.3 Calibration as distinguishability contraction

A central point for "non-ideal observers" is that calibration noise cannot improve distinguishability. For the Chernoff coefficient, this is a data-processing statement.

Theorem 2 (Calibration degrades Chernoff distinguishability)**.** *Let Λ be a CPTP map and ρ, σ quantum states. For every $s \in [0, 1]$,*

$$\operatorname{Tr}\left[\Lambda(\rho)^s \Lambda(\sigma)^{1-s}\right] \;\geq\; \operatorname{Tr}\left[\rho^s \sigma^{1-s}\right].$$

Equivalently, $Q(\Lambda(\rho), \Lambda(\sigma)) \geq Q(\rho, \sigma)$ and therefore

$$\xi(\Lambda(\rho), \Lambda(\sigma)) \leq \xi(\rho, \sigma).$$

Proof. For $s \in (0,1)$, define the Petz-Rényi divergence $D_s(\rho\|\sigma) = \frac{1}{s-1}\log \mathrm{Tr}[\rho^s\sigma^{1-s}]$. For $s \in (0,1)$ this divergence satisfies the data-processing inequality $D_s(\Lambda(\rho)\|\Lambda(\sigma)) \le D_s(\rho\|\sigma)$ under CPTP maps [17]. Rearranging yields the claimed inequality for $\mathrm{Tr}[\cdot]$. The endpoint cases $s = 0, 1$ follow by continuity. □

5 ε-SBS robustness of decision rules

The point of an explicit trace-distance SBS approximation is that it immediately controls *any* decision procedure via data processing.

Lemma 1 (Decision-theoretic continuity under trace distance)**.** *Let ρ and σ be two states on the same Hilbert space and let* Dec *be any (possibly adaptive) measurement-plus-decision procedure outputting a discrete hypothesis $\widehat{X}$. Then*

$$\left|\mathbb{P}_\rho(\widehat{X} \neq X) - \mathbb{P}_\sigma(\widehat{X} \neq X)\right| \le D_{\mathrm{tr}}(\rho, \sigma).$$

Proof. A measurement-plus-decision procedure is a CPTP map from states to a classical distribution on $\widehat{X}$. Trace distance contracts under CPTP maps, so the total variation distance between the induced classical distributions is bounded by $D_{\mathrm{tr}}(\rho, \sigma)$ [13]. The error event $\{\widehat{X} \neq X\}$ is a measurable subset of outcomes, hence its probability differs by at most the total variation distance. □

Theorem 3 (ε-robust Chernoff sample complexity)**.** *Let $\rho_{S\mathcal{E}}$ be ε-SBS with witness $\sigma_{S\mathcal{E}}$. Fix a fragment set A and an observer O. Let $\delta \in (0, \frac{1}{2})$ be a target error under the* actual *state ρ.*

If $\xi_O(A) \ge \log\left(\frac{1}{2(\delta-\varepsilon)}\right)$ and $\delta > \varepsilon$, then there exists a decision procedure on A (namely the optimal decoder for the witness experiment) such that

$$\mathbb{P}_\rho(\widehat{X} \neq X) \le \delta.$$

Proof. Apply Proposition 1 to the witness state σ: with the given $\xi_O(A)$, there exists a decoder achieving $\mathbb{P}_\sigma(\widehat{X} \neq X) \le \delta - \varepsilon$. Then Lemma 1 gives $\mathbb{P}_\rho(\widehat{X} \neq X) \le (\delta - \varepsilon) + \varepsilon = \delta$. □

6 Local versus collective decoding: a sharp gap for pure records

Observer temporal limitations often restrict *which measurements are feasible*. A key example is the distinction between collective measurements across m fragments (requiring quantum memory for coherence) and product measurements (measuring each fragment immediately).

For two pure states $|\psi_0\rangle, |\psi_1\rangle$ with overlap $c := |\langle\psi_0\rangle\psi_1| \in (0,1)$, the m-copy conditional states are $|\psi_0\rangle^{\otimes m}$ and $|\psi_1\rangle^{\otimes m}$ with overlap c^m. The quantum Chernoff coefficient is c^{2m}, giving exponent $-\log c^2$ per copy.

By contrast, any measurement on a *single copy* induces classical distributions p, q whose Bhattacharyya coefficient $B(p,q) = \sum_y \sqrt{p(y)q(y)}$ is bounded below by the square-root fidelity [18, 19]. For pure states this yields $B(p,q) \ge c$. For product measurements on m copies, the classical Chernoff coefficient is at least c^m, yielding exponent at most $-\log c$ per copy.

Corollary 1 (A factor-of-two exponent gap for pure-state records)**.** *Let $|\psi_0\rangle, |\psi_1\rangle$ be pure states with overlap $c \in (0,1)$. Let $P_e^{\mathrm{coll}}(m)$ be the optimal Bayes error using* collective *measurements on m*

copies, and let $P_e^{\mathrm{prod}}(m)$ be the optimal Bayes error over all product *measurements (local POVMs on each copy, followed by optimal classical postprocessing). Then the optimal error exponents satisfy*

$$\lim_{m\to\infty} -\frac{1}{m}\log P_e^{\mathrm{coll}}(m) = -\log(c^2), \qquad \limsup_{m\to\infty} -\frac{1}{m}\log P_e^{\mathrm{prod}}(m) \le -\log(c).$$

In particular, the best achievable exponent under product measurements is at most half the collective-measurement exponent.

Proof. The collective-measurement exponent is the quantum Chernoff exponent for pure states, which equals $-\log(c^2)$ [14, 15]. For product measurements, first measure each copy with some POVM, inducing classical distributions with Bhattacharyya coefficient $B \ge c$ [18, 19]. For i.i.d. classical samples, the classical Chernoff coefficient equals $\min_{s\in[0,1]} \sum_y p(y)^s q(y)^{1-s} \ge B$ (since $s = 1/2$ is feasible and gives B). Thus the m-sample coefficient is at least c^m, giving exponent at most $-\log c$. □

Remark 1 (Interpretation for observer quality)**.** If τ_O is too small to maintain coherence across many fragments, then the observer is effectively restricted to product measurements, reducing the achievable redundancy by a *quantifiable* amount (Cor. 1). This provides a mathematically explicit route from temporal observer constraints to reduced objectivity.

7 Worked physical model: central-spin pure dephasing

We now provide a concrete open-system model and compute (R_O, C_O, τ_O) in terms of physical parameters.

7.1 Model

Let S be a qubit with pointer basis the eigenbasis of σ_z^S. Let the environment consist of N qubits with $\sigma_z^{(k)}$. Consider the pure-dephasing Hamiltonian

$$H = \sigma_z^S \otimes \sum_{k=1}^{N} g_k \sigma_z^{(k)}, \tag{5}$$

and initial environment state $|+\rangle^{\otimes N}$ with $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Conditioned on the pointer value $x \in \{0,1\}$ (eigenvalues ± 1 of σ_z^S), fragment k evolves to a pure state

$$|\phi_k^{(x)}(t)\rangle = e^{\mp i g_k t \sigma_z} |+\rangle ,$$

so that the single-fragment overlap is

$$c_k(t) := \left| \langle \phi_k^{(0)}(t)\rangle \, \phi_k^{(1)}(t) \right| = |\cos(2 g_k t)| . \tag{6}$$

This model (and its variants) is standard in decoherence theory and is directly connected to SBS formation in spin environments [5, 20].

7.2 Exact Chernoff exponent and fragment requirement

For pure states, the Chernoff coefficient equals the fidelity:

$$Q\left(|\phi_k^{(0)}(t)\rangle\,,|\phi_k^{(1)}(t)\rangle\right) = c_k(t)^2, \qquad \xi_k(t) = -\log c_k(t)^2.$$

For a set A of accessed qubits,

$$\xi(A;t) = \sum_{k\in A} -\log\left(\cos^2(2g_k t)\right), \tag{7}$$

and Theorem 1 implies

$$P_e^\star(A;t) \le \tfrac{1}{2}\exp(-\xi(A;t))\,.$$

Thus a sufficient condition for error $\le \delta$ is $\xi(A;t) \ge \log(1/(2\delta))$.

7.3 Mapping to observer parameters

Access fraction R_O. In a laboratory implementation, an observer may only intercept a fraction of environment qubits (geometric coverage, detector efficiency, spatial range). If $m_{\max}$ qubits are accessible per trial out of N, then $R_O = m_{\max}/N$.

Calibration C_O. Suppose the observer's instrument applies a depolarizing channel on each accessed environment qubit: $\Lambda_p(\rho) = (1-p)\rho + p\frac{\mathbf{1}}{2}$, capturing miscalibration or noise before readout. For any two qubit states, depolarization contracts trace distance by a factor $(1-p)$ [13]. For the central-spin conditional pure states, the single-copy Helstrom success probability becomes

$$C_O(t) = \tfrac{1}{2}\left(1 + (1-p)\sqrt{1-c_k(t)^2}\right) = \tfrac{1}{2}\left(1 + (1-p)|\sin(2g_k t)|\right),$$

where we used $c_k(t) = |\cos(2g_k t)|$. If couplings differ, $C_O(t)$ varies with k and can be summarized by a worst-case or average calibration across accessible fragments.

Temporal horizon τ_O. If the observer performs collective decoding across m fragments, a quantum memory must preserve coherence for at least the acquisition time of those m qubits plus the decoding time. Let t_{meas} be the per-qubit acquisition time. A minimal feasibility constraint is $\tau_O \gtrsim m\, t_{\text{meas}}$. Additionally, the pointer value must remain stable over the acquisition interval, typically limited by system relaxation times (e.g., T_1) [21, 22]. In this model, we treat stability as an explicit constraint rather than imposing an unjustified "exponential stability" assumption.

7.4 Numerical illustration and reproducibility

The accompanying repository includes a script that samples couplings $\{g_k\}$, computes $\xi(A;t)$ for the best accessible subset under an access fraction R_O, and plots redundancy as a function of time t. This is included to make the "worked example" fully reproducible. Figure 1 shows a typical redundancy profile as a function of interaction time in this toy model.

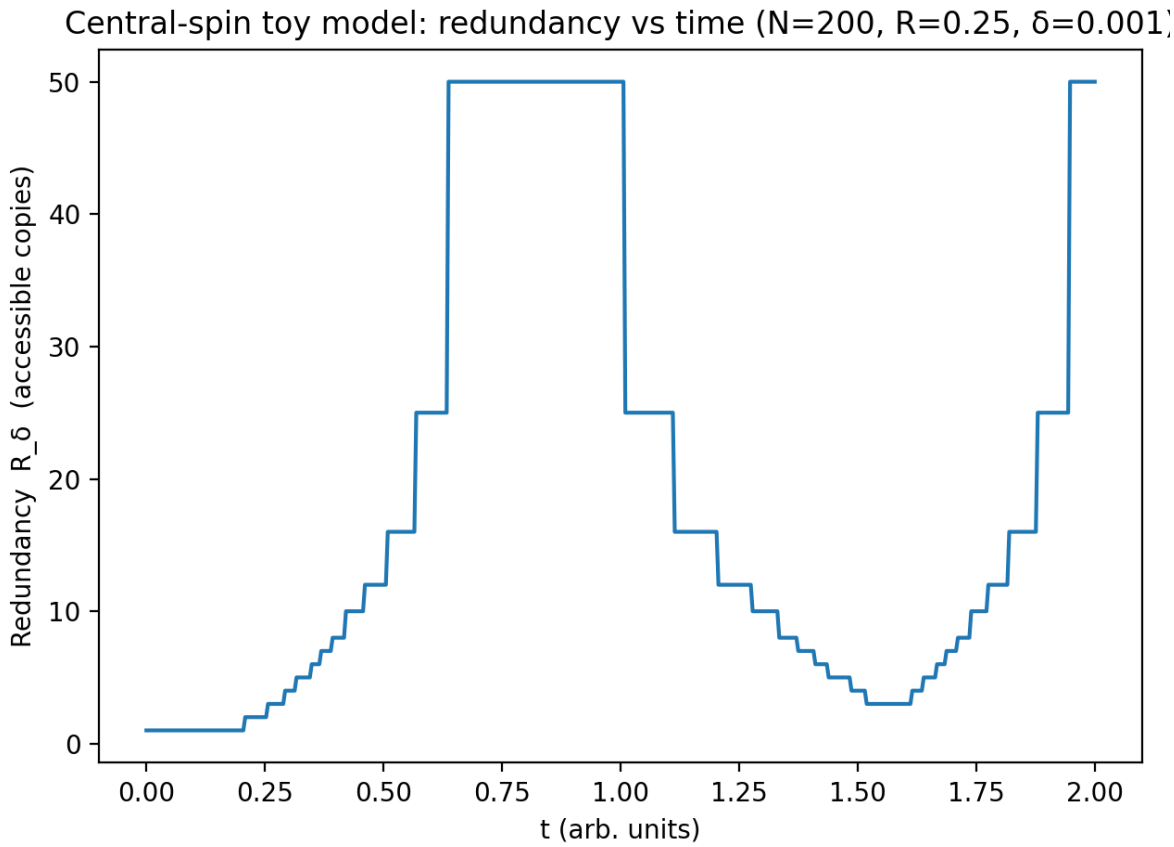

Figure 1: Central-spin toy model (Sec. 7): redundancy (accessible copies) versus interaction time t for a sampled distribution of couplings $\{g_k\}$, access fraction $R_O = 0.25$, and target decoding error $\delta = 10^{-3}$. The curve is produced by the reproducibility script in the repository.

8 DLN-series connection (brief)

This paper is part of a DLN series in which observers are modeled as bounded information-processing systems. For the purposes of a physics submission, we keep the DLN connection minimal: the DLN compression model provides an *independent* motivation for parameterizing observer limitations via access, calibration, and time horizons (resources) [23, 24]. The empirical/cognitive design component is planned as a separate paper (4B) and is therefore out of scope here.

9 Discussion: tightness, limitations, and relation to prior work

We highlight three points relevant for interpreting the bounds and comparing to prior work:

- **Mathematical depth:** the main redundancy bound is phrased in terms of Chernoff-optimal decoding [14, 15] and explicitly incorporates calibration via a CPTP contraction theorem (Thm. 2) and ε-SBS perturbations (Thm. 3). The local-vs-collective decoding gap (Cor. 1) links temporal constraints to a provable reduction in error exponent.
- **Prior work:** we explicitly compare to hazy environments [7], photon redundancy [10, 11], generic objectivity [8, 9], and accessible-information approaches [16].

- **Concrete physical model:** Sec. 7 gives a full worked open-system example computing fragment overlaps and Chernoff exponents from Hamiltonian parameters.
- **DLN connection:** condensed to Sec. 8.

10 Conclusion

Observer modeling is a structural gap in much of the QD/SBS literature. By treating the observer as a resource-constrained agent with explicit access, calibration, and temporal horizons, we obtain quantitative, decoder-level statements about redundancy and objectivity that remain stable under ε-SBS trace-distance approximations. The central-spin worked example demonstrates how these parameters can be computed from physical couplings and instrument noise, enabling model-to-experiment predictions. The factor-of-two exponent gap between local and collective decoding provides a sharp and testable link between temporal observer constraints and the emergence of objectivity.

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